REE Mathematics: Linear Systems, Complex Models, and Data

REGISTERED-ELECTRICAL-ENGINEERING · registered-electrical-engineering-mathematics · BoardPrep lesson v1

Official examination scope

PRC’s current REE TOS and examination program guide subject scope. This original module connects selected algebra, calculus, phasor, transform, probability, and economic methods; it is not a complete REE mathematics course.

Learning objectives

- Solve a simultaneous linear model and verify every original equation.
- Relate derivatives and differential equations to change over time.
- Explain when complex and transform methods represent periodic or dynamic behavior.

1. Linear equations, matrices, and determinants

A system of linear equations represents several constraints on the same unknowns. Elimination, substitution, or matrix methods can solve the system when the coefficient structure supports a unique solution. After calculating, substitute the result into every original equation; a check can reveal arithmetic or transcription errors. A determinant helps diagnose invertibility for a square matrix, but numerical conditioning matters when inputs are approximate. In an engineering interpretation, variables retain their units and the equations must represent compatible physical relationships.

2. Calculus, rates, and differential equations

A derivative models instantaneous change with respect to a defined variable; its units are the dependent quantity’s units divided by the independent quantity’s units. Integration accumulates a rate over an interval, subject to bounds and initial conditions. Differential equations express how a system’s state changes and require conditions that identify a particular solution. Interpret an equation alongside the physical meaning of its variables and sign convention; a mathematically correct result can describe the wrong circuit or boundary condition if the model was set up incorrectly.

3. Complex numbers, phasors, Fourier, and Laplace methods

Complex numbers combine real and imaginary components; magnitude and angle offer a useful representation for sinusoidal steady-state quantities. A phasor is tied to a common frequency and convention, and it does not describe arbitrary transient behavior by itself. Fourier methods represent periodic content through frequency components, while Laplace transforms can solve linear differential systems with specified initial conditions. Keep transform variables, units, and initial-value terms visible. Check whether the problem asks for steady state, transient response, or a periodic decomposition before selecting a method.

4. Probability, geometry, and economic comparisons

Probability and statistics quantify uncertainty using a defined sample space, data collection process, and model. Check assumptions before applying a distribution or interpreting a confidence measure. Trigonometry and analytic geometry connect angles, lengths, and coordinates; label the reference angle and units. Engineering-economics calculations compare alternatives using timing, interest, service horizon, and stated costs. A present-worth result is conditional on the inputs and does not make a safety or ethical tradeoff automatically acceptable.

Solve a two-equation current model

Question: A feeder model gives 2x + y = 11 and x - y = 1, where x and y are branch currents in amperes. Find x.
1. Add the equations to eliminate y: 3x = 12.
2. Divide by 3 to get x = 4 A.
3. Substitute into x - y = 1 to obtain y = 3 A, then verify 2(4) + 3 = 11.
Answer: x = 4 A.
Check: Both unknowns satisfy both original equations.

Common misconceptions

- A solved equation system is correct without substitution.
- A phasor represents every time-varying waveform.
- More displayed digits mean greater measurement accuracy.

Practice patterns

- Check every original constraint after solving.
- Match the mathematical method to the time behavior.
- Name model, units, and assumptions for data or economic analysis.

Official scope references

Official scope: Enhanced Tables of Specifications for REE — Annex A — Professional Regulatory Board of Electrical Engineering, Professional Regulation Commission — https://www.prc.gov.ph/sites/default/files/2024-40%20Annex%20A.pdf
Official scope: Program of the September 2026 Registered Electrical Engineers Licensure Examination — Professional Regulation Commission; Professional Regulatory Board of Electrical Engineering — https://www.prc.gov.ph/sites/default/files/Program%20September%202026%20Electrical%20Engineers.pdf

Instructional references

Instructional reference: Precalculus 2e — OpenStax, Rice University — https://openstax.org/details/books/precalculus-2e
Instructional reference: Calculus Volume 1 — OpenStax, Rice University — https://openstax.org/details/books/calculus-volume-1
Instructional reference: MIT OpenCourseWare 18.03SC — Fourier Series and Laplace Transform — Massachusetts Institute of Technology OpenCourseWare — https://ocw.mit.edu/courses/18-03sc-differential-equations-fall-2011/pages/unit-iii-fourier-series-and-laplace-transform/
Instructional reference: NIST/SEMATECH Engineering Statistics Handbook — National Institute of Standards and Technology; SEMATECH — https://www.nist.gov/programs-projects/nistsematech-engineering-statistics-handbook

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